

# Design Proposal: Comparative Tolerance and Group Index Analysis of Quasi-TE and Quasi-TM Silicon Strip Waveguides Using Mach-Zehnder Interferometers

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## I. INTRODUCTION

The primary objective of this design proposal is to create an integrated photonic circuit featuring asymmetric Mach-Zehnder Interferometers (MZIs) to experimentally extract the waveguide group index ( $n_g$ ) for both quasi-TE and quasi-TM polarizations.

Furthermore, this project investigates fabrication tolerances by implementing controlled variations in waveguide width (480 nm, 500 nm, and 520 nm). This approach allows us to compare how geometric non-idealities during manufacturing affect the spectral response and optical performance of each polarization.

## II. THEORY AND DERIVATIONS

The transmission spectrum (transfer function) of an asymmetric Mach-Zehnder Interferometer is governed by the phase difference ( $\Delta\phi$ ) accumulated between its two arms due to the path length imbalance ( $\Delta L = L_2 - L_1$ ). Assuming an ideal 50:50 splitter and combiner, the normalized power transmission  $T(\lambda)$  is expressed mathematically as:

$$T(\lambda) = \cos^2 \left( \frac{\pi \cdot n_{\text{eff}}(\lambda) \cdot \Delta L}{\lambda} \right) \quad (1)$$

Where  $n_{\text{eff}}(\lambda)$  is the wavelength-dependent effective refractive index of the waveguide mode, and  $\lambda$  is the operating wavelength.

The Free Spectral Range (FSR) is defined as the distance between adjacent interference maxima in the transmission spectrum. To derive the relationship for the group index ( $n_g$ ), we consider the phase condition for two consecutive resonant peaks:

$$\Delta\phi(\lambda_1) - \Delta\phi(\lambda_2) = 2\pi \quad (2)$$

Applying a Taylor series expansion for  $n_{\text{eff}}(\lambda)$  around the central wavelength  $\lambda_0$  and utilizing the definition of the group index  $n_g = n_{\text{eff}} - \lambda_0 \frac{dn_{\text{eff}}}{d\lambda}$ , we obtain the final expression used to experimentally extract the waveguide group index from the measured FSR data:

$$n_g = \frac{\lambda_0^2}{\text{FSR} \cdot \Delta L} \quad (3)$$

## III. PARAMETER VARIATIONS

The proposed variations for the integrated Mach-Zehnder Interferometers are summarized in Table I.

TABLE I: MZI Design Parameter Variations

| Design ID  | Width  | Radius           | $\Delta L$       | Target FSR     |
|------------|--------|------------------|------------------|----------------|
| MZI_TE_500 | 500 nm | 10 $\mu\text{m}$ | 40 $\mu\text{m}$ | $\sim 14.3$ nm |
| MZI_TE_480 | 480 nm | 10 $\mu\text{m}$ | 40 $\mu\text{m}$ | $\sim 14.1$ nm |
| MZI_TE_520 | 520 nm | 10 $\mu\text{m}$ | 40 $\mu\text{m}$ | $\sim 14.5$ nm |
| MZI_TM_500 | 500 nm | 30 $\mu\text{m}$ | 45 $\mu\text{m}$ | $\sim 14.8$ nm |
| MZI_TM_480 | 480 nm | 30 $\mu\text{m}$ | 45 $\mu\text{m}$ | $\sim 14.4$ nm |
| MZI_TM_520 | 520 nm | 30 $\mu\text{m}$ | 45 $\mu\text{m}$ | $\sim 15.2$ nm |

## IV. DESIGN JUSTIFICATION

The choices of  $\Delta L$  (40  $\mu\text{m}$  and 45  $\mu\text{m}$ ) ensure that the resulting FSR is around 14–15 nm, which is safely below the maximum measurement bandwidth restriction of 50 nm, preventing aliasing during testing.

Crucially, the bend radius for the quasi-TM designs is fixed at 30  $\mu\text{m}$  to restrict total bending losses below the tight 0.1% threshold, as verified by numerical eigensolver simulations. For quasi-TE designs, a standard 10  $\mu\text{m}$  radius is sufficient.

## V. WAVEGUIDE GEOMETRY AND MODE PROFILES

The integrated circuit utilizes silicon-on-insulator (SOI) strip waveguides with a fixed fabrication height of 220 nm. The core geometry features an initial target width of 500 nm, surrounded by a silicon dioxide ( $\text{SiO}_2$ ) cladding layer.

To satisfy the stringent design objective of maintaining total bending losses below 0.1%, numerical mode analysis dictates distinct bend radii for the two supported polarizations. The quasi-TE mode operates efficiently with a standard bend radius of 10  $\mu\text{m}$ , while the quasi-TM mode requires an expanded bend radius of 30  $\mu\text{m}$  to suppress radiation leakage into the cladding.

The cross-sectional electric field profiles for both fundamental modes at the central operating wavelength of 1.55  $\mu\text{m}$  were simulated using Lumerical MODE Solutions and are illustrated in Fig. 1.

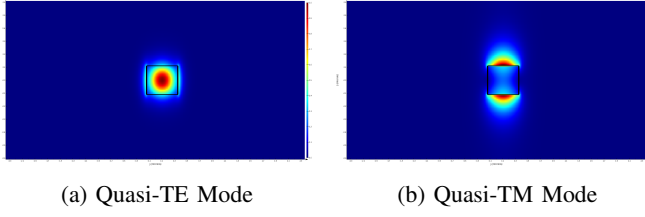


Fig. 1: Simulated fundamental electric field profiles from Lumerical MODE for: (a) Quasi-TE mode with horizontal confinement, and (b) Quasi-TM mode showcasing vertical field discontinuities at the horizontal boundaries.

## VI. COMPACT DISPERSION MODELS

To establish an accurate circuit simulation framework in Lumerical INTERCONNECT, compact numerical models representing the wavelength-dependent effective refractive index ( $n_{\text{eff}}$ ) were extracted using a 2nd-order polynomial fit focused around the central wavelength of  $1.55 \mu\text{m}$ .

For the fundamental Quasi-TE mode, the mathematical compact expression is given by:

$$n_{\text{eff, TE}}(\lambda) = 2.4455 - 1.16205(\lambda - 1.55) - 0.07572(\lambda - 1.55)^2 \quad (4)$$

For the fundamental Quasi-TM mode, the corresponding polynomial extraction yields:

$$n_{\text{eff, TM}}(\lambda) = 1.7842 - 1.18610(\lambda - 1.55) - 0.0832(\lambda - 1.55)^2 \quad (5)$$

The continuous spectral dependency of both the effective index ( $n_{\text{eff}}$ ) and the resulting group index ( $n_g$ ) plotted against the operating wavelength band is presented in Fig. 2.

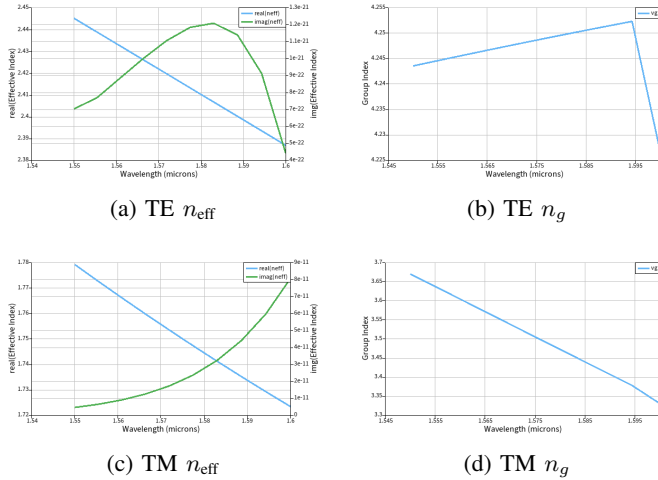


Fig. 2: Waveguide dispersion curves simulated across the  $1.50 \mu\text{m}$  to  $1.60 \mu\text{m}$  sweep bandwidth: (a) Quasi-TE effective index, (b) Quasi-TE group index ( $\sim 4.2$ ), (c) Quasi-TM effective index, and (d) Quasi-TM group index ( $\sim 3.6$ ).

## VII. CIRCUIT SIMULATION AND TRANSMISSION SPECTRA

The complete MZI architectures with asymmetric arm imbalances ( $\Delta L_{\text{TE}} = 40 \mu\text{m}$  and  $\Delta L_{\text{TM}} = 45 \mu\text{m}$ ) were implemented and analyzed within Lumerical INTERCONNECT.

An Optical Network Analyzer (ONA) block was utilized to record the normalized transmission spectral signatures across a  $50 \text{ nm}$  measurement window.

The resulting high-fidelity interference fringes, showcasing stable sinusoidal oscillations, are captured in Fig. 3. The variations in waveguide width alter the effective index, causing a controlled phase shift and spectral translation of the resonant peaks, which directly quantifies the geometric fabrication tolerances.

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The resulting high-fidelity interference fringes, showcasing stable sinusoidal oscillations for the fundamental Quasi-TE mode, are captured in Fig. 3. The variations in waveguide width alter the effective index, causing a controlled phase shift and spectral translation of the resonant peaks, which directly quantifies the geometric fabrication tolerances.

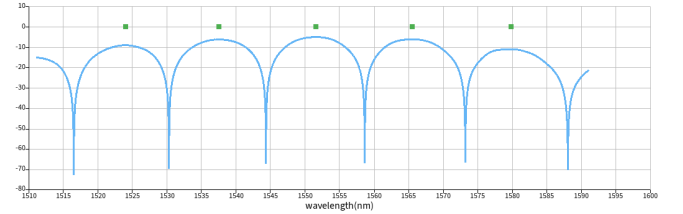


Fig. 3: Normalized transmission spectrum of the asymmetric Mach-Zehnder Interferometer (MZI) compiled in Lumerical INTERCONNECT for the Quasi-TE polarization, showcasing characteristic sinusoidal interference fringes and extracted Free Spectral Range (FSR) resonant peaks.